

Smart Loan Risk & Approval System

Business Report

End-to-End Machine Learning Application
for Loan Risk Prediction and Approval

Report Summary

Report Type:

Business Report

Generated On:

2026-09-28 21:08

Total Records:

5,000

Total Columns:

21

Target Column:

Loan_Status

Active Model:

Logistic Regression

Smart Loan Risk & Approval System

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1. Dataset Overview

Total Records	Total Columns	Numeric Cols	Categorical Cols
5,000	21	11	10

2. Data Quality

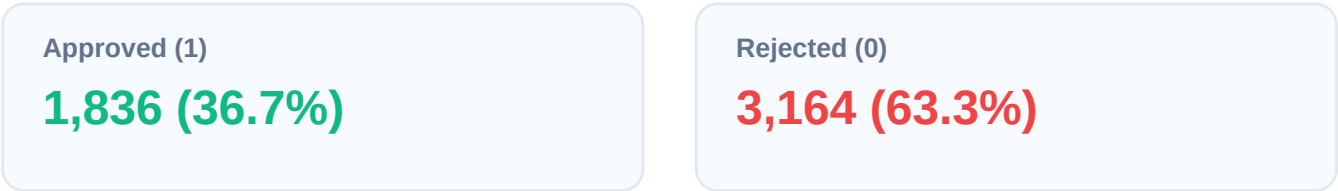
Metric	Count	Percentage
Missing Values	0	0.00%
Duplicate Rows	0	0.00%
Numerical Cols	11	52%
Categorical Cols	10	48%

Dataset is ready for analysis and model training.

3. Missing Values by Column

No missing values detected.

4. Target Analysis (Loan_Status)



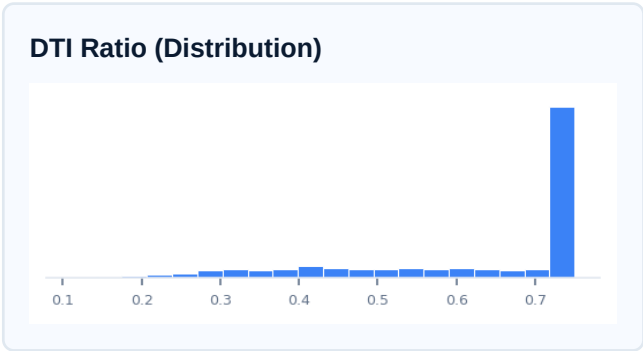
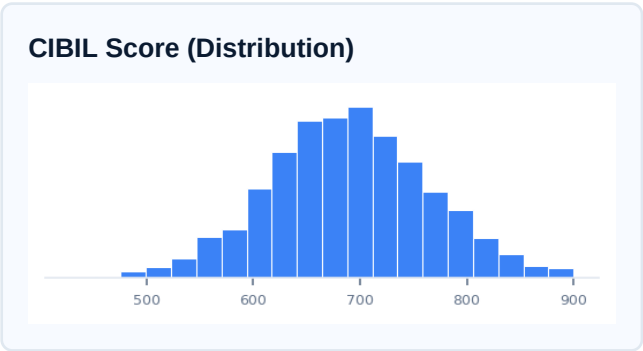
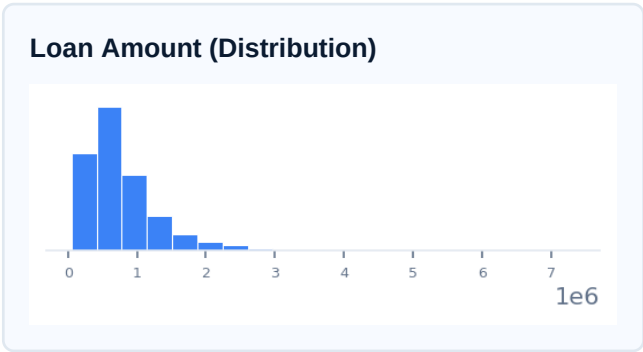
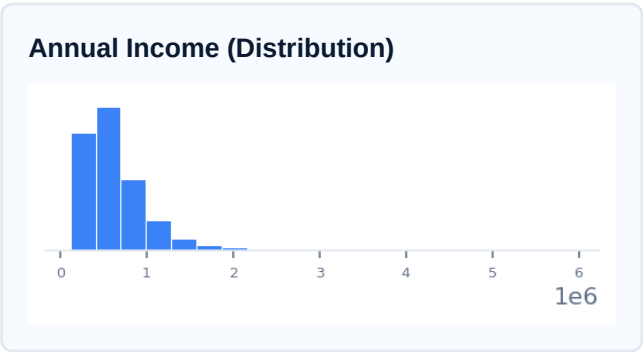
5. Descriptive Statistics

Feature	Mean	Median	Min	Max	Std Dev
Age	42.9	43.0	21.0	65.0	13.0
Employment_Years	15.7	15.1	-0.0	35.0	9.1
Annual_Income	630699.8	540000.0	120000.0	5942000.0	375636.2
Coapplicant_Income	102987.6	65000.0	0.0	1856000.0	140083.2
Loan_Amount	792643.4	638000.0	51000.0	7344000.0	588091.3
Loan_Term_Years	8.8	7.0	2.0	20.0	5.3
Cibil_Score	689.4	689.0	429.0	900.0	75.4
Existing_Loans	1.2	1.0	0.0	6.0	1.1
DTI_Ratio	0.6	0.8	0.1	0.8	0.2
Residential_Assets	2080798.6	1369500.0	66000.0	80846000.0	2521706.4
Bank_Assets	478127.0	343000.0	22000.0	7727000.0	458559.1

6. Outlier Analysis (IQR Method)

Feature	Outlier Count	Outlier %
Annual_Income	227	4.5%
Coapplicant_Income	228	4.6%
Loan_Amount	282	5.6%
Cibil_Score	43	0.9%
Residential_Assets	372	7.4%
Bank_Assets	309	6.2%

7. Distribution of Key Features



8. Model Training & Comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.767	0.786	0.854	0.819	0.841

Only one successfully evaluated model is available.

Selected Model: Logistic Regression

Model Performance Summary

The Logistic Regression model was selected based on optimal evaluation metrics. It achieved an accuracy of 76.7% on the validation set, demonstrating strong generalization capability for predicting loan risk.

9. Active Model Details

Model Name

Logistic Regression

Model Version

v20260928210715

Model Type

Loan Risk Classification

Training Status

Ready for Prediction

10. New Applicant Prediction

Applicant Information

No applicant data available.

Prediction Result

Prediction:
LOW RISK

Model Confidence: 85.0%

Approval Probability: 85.0%

Risk Probability: 15.0%

Decision: **APPROVE**

11. Final Summary

The system analyzed 5,000 historical loan applications using Logistic Regression. The selected model achieved 76.7% accuracy on the evaluation dataset. The latest applicant was classified as LOW RISK with 85.0% confidence.